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Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

NATIONAL SENIOR CERTIFICATE/NASIONALE SENIOR SERTIFIKAAT

GRADE/GRAAD 12

**MATHEMATICAL LITERACY P2/
WISKUNDIGE GELETTERDHEID V2**

NOVEMBER 2021

MARKING GUIDELINES/NASIENRIGLYNE

MARKS/PUNTE: 150

SYMBOL/KODE	EXPLANATION/VERDUIDELIKING
M	Method/Metode
MA	Method with accuracy/Metode met akkuraatheid
CA	Consistent accuracy/Volgehoue akkuraatheid
A	Accuracy/Akkuraatheid
C	Conversion/Herleiding
S	Simplification/Vereenvoudiging
RT	Reading from a table/graph/map/diagram/Lees vanaftabel/kaart/grafiek/diagram
SF	Correct substitution in a formula/Korrekte vervanging in formule
O	Opinion/Explanation/Reasoning /Opinie/Verduideliking/Redenasie
P	Penalty, e.g. for no units, incorrect rounding off, etc./Penalisering, bv. vir geen eenhede/verkeerde afronding, ens.
R	Rounding off/Afronding
NPR	No penalty for correct rounding minimum two decimal places/Geenpenaliseringvir korrekte afronding tot twee desimale plekke nie
AO	Answer only/Slegs antwoord
MCA	Method with constant accuracy/Metode met volgehoue akkuraatheid

**These marking guidelines consist of 19 pages.
Hierdienasienriglynebestaanuit 19 bladsye.**

NOTE:

- If a candidate answers a question TWICE, only mark the FIRST attempt.
- If a candidate has crossed out (cancelled) an attempt to a question and NOT redone the solution, mark the crossed out (cancelled) version.
- Consistent accuracy (CA) applies in ALL aspects of the marking guidelines; however it stops at the second calculation error.
- Note: consistent accuracy (CA) does not apply in cases of a breakdown.
- If the candidate presents any extra solution when reading from a graph, table, layout plan and map, then penalise for every extra item presented.

As a general marking principle, if a candidate has incurred one mistake and there is evidence of sound mathematics thereafter, then that candidate should lose one mark only.

LET WEL:

- As 'n kandidaat 'n vraag TWEE KEER beantwoord, merk slegs die EERSTE poging.
- As 'n kandidaat 'n antwoord van 'n vraag doodtrek (kanselleer) en nie oordoen nie, merk die doodgetrekte (gekanselleerde) poging.
- Volgehoue akkuraatheid (CA) word in ALLE aspekte van die nasienriglyne toegepas, dit hou op by die tweede berekeningsfout.
- Let wel: volgehoue akkuraatheid (CA) geld nie in die geval van 'n afbreuk nie.
- Wanneer 'n kandidaat aflesings vanaf 'n grafiek, tabel, uitlegplan en kaart geneem en ekstra antwoorde gee, penaliseer vir elke ekstra item.
- 'n Algemene merkbeginsel is dat indien 'n kandidaat een fout maak en daarna voortgaan met korrekte wiskunde, dat die kandidaat slegs een punt verloor.

QUESTION/VRAAG 1 [29 MARKS/PUNTE] Answer Only AO - full marks			
Q/V	Solution/Oplossing	Explanation/Verduideliking	T/L
1.1.1	Total mass/Totale massa = $6 \times 110\text{g}$ ✓MA = 660 g ✓A	1MA multiply mass by 6 1A mass (2)	M L1
1.1.2*	Radius = 32 mm ✓✓A	2A radius (2)	M L1
1.1.3	A ✓✓A	2A correct letter [accept: mm ³] (2)	M L1
1.1.4*	Total No. of days/Totale getal dae = 11 Jan to 31 Mar ✓ MA = $(31 - 10) + 28 + 31$ ✓ MCA = $21 + 28 + 31 = 80$ ✓CA	1MA days in Jan 1MCA adding days in 3 months 1CA simplification (3)	M L1

Q/V	Solution/Oplossing	Explanation/Verduideliking	T/L
1.1.5*	Price for 2 Pringles/Prys vir 2 Pringles $= 2 \left(\frac{R100}{6} \right)$ ✓ MA $= 2 \times R16,666$ $= R33,33$ ✓ CA	1MA dividing price by 6 and multiplying by 2 1CA simplification NPR (2)	M/F L1
1.2.1	A ✓✓ A	2A correct letter (2)	M L1
1.2.2	D ✓✓ A	2A correct letter Accept 60 km/h (2)	M L1
1.3.1	$7,3 \text{ m} = 7,3 \times 100 \text{ cm}$ ✓ MA $= 730 \text{ cm}$ ✓ A	1MA multiplying correct value by 100 1A simplification (2)	M L1
1.3.2*	$D = 7,3 \text{ m} - 5,2 \text{ m}$ ✓ MA $= 2,1 \text{ m}$ ✓ CA	1MA difference of correct lengths 1CA simplification (2)	M L1
1.3.3	0,5m ✓✓ A	2A height (2)	M L1
1.4.1*	✓A A layout plan is a top view that shows the arrangement of features. ✓A <i>'n Uitlegplan is die bo-aansig wat die rangskikking van die voorwerpe aantoon.</i> OR/OF A layout plan is the structural arrangement of items within a certain space. <i>'n Uitlegplan is die strukturele rangskikking van items binne 'n bepaalde ruimte.</i> OR/OF Plan of the entire inside cabin, showing location of seats, exit doors etc. <i>'n Plan van die hele binnekant van die kajuit wat die posisie van sitplekke, uitgang, deure ens. aantoon</i> OR/OF Drawing to scale showing physical arrangements of all resources that consume space within facilities. <i>'n Skaaltekening wat die fisiese posisies van al die items van spasie in beslag neem binne die fasiliteit</i>	2A explanation (2)	MP L1

Q/V	Solution/Oplossing	Explanation/Verduideliking	T/L
1.4.2*	28 ✓✓ A	2A number of seats (2)	MP L1
1.4.3	✓ A ✓ A G1	1A correct seat 1A correct row (2)	MP L1
1.4.4*	6 ✓✓ A	2A correct number (2)	P L1
		[29]	

QUESTION/VRAAG 2 [24MARKS/PUNTE]			
Q/V	Solution/Oplossing	Explanation/Verduideliking	T/L
2.1	3 ✓✓A	2A correct number (2)	MP L2
2.2	Living room/Woonkamer ✓✓A	2A correct room (2)	MP L1
2.3	North East or NE/Noordoos of NO ✓✓A	2A direction (2)	MP L2
2.4*	$P_{\text{not interior/nie binne}} = P_{\text{exterior/buite}}$ $= \frac{2}{6} \quad \checkmark\checkmark\text{RT}$ $= \frac{1}{3} \quad \checkmark\text{A}$ $= \frac{1}{3} \quad \checkmark\text{CA}$ OR/OF $P_{\text{not interior/nie binne}} = 1 - \frac{4}{6} \quad \checkmark\text{MA}$ $= \frac{2}{6} \quad \checkmark\text{RT}$ $= \frac{1}{3} \quad \checkmark\text{A}$ $= \frac{1}{3} \quad \checkmark\text{CA}$	2RT numerator 1A denominator 1CA simplification OR/OF 1MA probability of NOT 1RT numerator 1A denominator 1CA simplification (4)	P L2
2.5	✓A ✓✓O Jan is wrong, the kitchen is on the Southern side. In South Africa it does not get a lot of sun. <i>Jan is verkeerd. Die kombuis is aan die suidlike kant. In Suid-Afrika kry dit nie baie son nie.</i>	1A wrong 2O reasoning (3)	MP L4
2.6	It cannot be the view showing the kitchen and dining room, as it does not show the extra window for the bathroom. ✓✓O <i>Dit kannie die kombuis en eetkamer wees nie want dit wys nie die venster van die badkamer nie.</i> It does not show the other rooms on both sides of the windows. <i>Dit wys nie die ander kamers weerskante van die vensters nie.</i> OR/OF It shows the veranda, door, bedroom and livingroom windows. <i>Dit wys die stoep, deur en slaapkamer en woonkamervensters.</i> OR/OF	2O reason	MP L4

Q/V	Solution/Oplossing	Explanation/Verduideliking	T/L
	OR/OF Because there is no veranda on the side of the kitchen and the picture shows the veranda. <i>Daar is nie 'n stoep aan die kombuis se kant nie en die prent toon 'n stoep</i> OR/OF The drawing shows the SE elevation and the kitchen is on the SW side. <i>Die prent toon die SO aansien die kombuis is aan die SW kant.</i> OR/OF The windows shown does not look like kitchen windows, they are too big. <i>Die vensters wat aangetoon word lyk nie soos kombuisvensters nie, hulle is te groot</i> OR/OF The drawing represents the front view. <i>Die prent is die vooraansig</i> OR/OF Kitchen should be on the left-hand side with the window and door / The door knob is on the right-hand side and not on the left-hand side of the door adjacent to the kitchen window. <i>Kombuis moet aan die linkerkant met die vensterendeur wees / Die deurknop is aan die regterkant en nie aan die linkerkant van die deur wat grens aan die kombuisvenster nie.</i>		
2.7.1*	10 mm : 1 000 mm ✓A = 1 : 100 ✓CA OR/OF 1 cm : 100 cm ✓A = 1 : 100 ✓CA	1A correct ratio and conversion 1CA simplification OR/OF 1A correct ratio and conversion 1CA simplification AO	MP L2
2.7.2	Length on floor plan/Lengte op die vloerplan = 4,4 cm ✓A 1 cm = 100 cm 4,4 cm = 4,4 × 100 cm ✓MCA = 440 cm ✓CA = 4,4 m ✓C OR/OF	CA from 2.7.1 1A correct measurement 1MCA using the scale 1CA simplification 1C conversion Accept 4,3 m to 4,5 m	MP L3

Q/V	Solution/Oplossing	Explanation/Verduideliking	T/L
	1 cm is 1 000 mm ✓A 4,4 cm is 4 400 mm ✓MCA ✓CA 4 400 mm = 4,4 m ✓C OR/OF 1cm : 1 000 mm ✓MCA 1cm : 1 m ✓C ✓A 4,4 cm : 4,4 m ✓CA	1A correct measurement 1MCA using the scale 1CA simplification 1C conversion OR/OF 1MCA using the scale 1C conversion 1A correct measurement 1CA simplification (4)	
2.7.3	Jan is correct. ✓A ✓✓O When a photocopy is made the size of the plan may change while the number scale remains the same. <i>Jan is korrek. Wanneer jy 'n fotostaat maak, kan die grootte van die plan verander en die getalskaal bly dieselfde</i>	1A opinion 2O verification (3)	MP L4
		[24]	

QUESTION/VRAAG 3 [35 MARKS/PUNTE]			
Q/V	Solution/Oplossing	Explanation/Verduideliking	T/L
3.1.1	$\begin{aligned} & \checkmark\text{RT} \qquad \checkmark\text{MA} \\ \mathbf{A} &= 162 \text{ cm} + 1,5 \text{ cm} + 1,5 \text{ cm} \\ &= 162 \text{ cm} + (1,5 \text{ cm} \times 2) \\ &= 165 \text{ cm} \quad \checkmark\text{CA} \end{aligned}$	1RT inside length 1MA adding both sides 1CA simplification (3)	M L1
3.1.2*	$\begin{aligned} & \checkmark\text{RT} \qquad \checkmark\text{MA} \\ \mathbf{B} &= 80 \text{ cm} - (40 \text{ cm} + 4,5 \text{ cm} + 1,5 \text{ cm} + 1,5 \text{ cm}) \\ &= 32,5 \text{ cm} \quad \checkmark\text{CA} \end{aligned}$	1RT both heights 1MA subtracting 1CA simplification (3)	M L1
3.2	$\begin{aligned} 31,496 \text{ inches/duim} &= 80 \text{ cm} \quad \checkmark\text{RT} \\ 1 \text{ inch/duim} &= \frac{80}{31,496} \text{ cm} \quad \checkmark\text{MA} \\ &= 2,54 \text{ cm} \quad \checkmark\text{A} \end{aligned}$	1RTheight 80 cm 1MA dividing by 31,496 1A simplification (3)	M L2
3.3.1	$\begin{aligned} \text{Area of a rectangle} &= \text{length} \times \text{width} \\ \text{Opp van 'n reghoek} &= \text{lengte} \times \text{breedte} \\ &= 165 \text{ cm} \times 80 \text{ cm} \quad \checkmark\text{MCA} \\ &= 13\,200 \text{ cm}^2 \quad \checkmark\text{CA} \end{aligned}$	CA from 3.1.1 1MCA substitution 1CA simplification (2)	M L2
3.3.2*	$\begin{aligned} \text{Area of a rectangle} &= 13\,200 \text{ cm}^2 \\ &= \frac{13200}{(100)^2} \text{ m}^2 \quad \checkmark\text{MCA} \\ &= 1,32 \text{ m}^2 \quad \checkmark\text{CA} \end{aligned}$ or Area $= 1,65 \times 0,8$ $= 1,32 \text{ m}^2$	CA from 3.3.1 1MCA dividing by 100^2 or 10 000 1CA simplification AO (2)	M L2
3.3.3	$\begin{aligned} 1 \ell \text{ covers/bedek} & 6,9 \text{ m}^2 \\ n \ell \text{ covers/bedek} & 1,32 \text{ m}^2 \\ n &= \frac{1,32}{6,9} \quad \checkmark\text{MA} \\ &= 0,1913... \ell \quad \checkmark\text{CA} \\ \text{To paint three coats/ Om drie lae te verf} & \\ & \checkmark\text{MA} \\ 0,1913... \ell \times 3 &= 0,57 \ell \quad \checkmark\text{CA} \\ & \checkmark\text{R} \end{aligned}$	CA from 3.3.2 1MA dividing by 6,9 1CA simplification 1MA multiplying with 3 1CA simplification 1R rounding	M L3

Q/V	Solution/Oplissing	Explanation/Verduideliking	T/L
	OR/OF Total area to cover / <i>Totale oppervlakte om te dek</i> $\checkmark\text{MA}$ $= 1,32 \text{ m}^2 \times 3 = 3,96 \text{ m}^2 \quad \checkmark\text{CA}$ 1 ℓ covers/<i>bedek</i> $6,9 \text{ m}^2$ $x \text{ ℓ covers /bedek } 3,96 \text{ m}^2$ $\checkmark\text{MA}$ $x = \frac{3,96}{6,9} = 0,57 \text{ ℓ} \quad \checkmark\text{CA} \quad \checkmark\text{R}$ OR/OF Paint needed/<i>Verf benodig</i> $\checkmark\text{MA}$ $= \frac{1,32 \times 2}{6,9} \text{ ℓ} + \frac{1,32}{6,9} \text{ ℓ} \quad \checkmark\text{MA}$ $\checkmark\text{CA} \quad \checkmark\text{CA}$ $= 0,38 \text{ ℓ} + 0,19 \text{ ℓ}$ $= 0,57 \text{ ℓ} \quad \checkmark\text{R}$ OR/OF Total area to cover / <i>Totale oppervlakte om te dek</i> $\checkmark\text{MA} \quad \checkmark\text{CA}$ $= 1,32 \text{ m}^2 \times 3 = 3,96 \text{ m}^2$ Spread rate/ <i>Spreikoers</i> = $\frac{1 \text{ ℓ}}{6,9 \text{ m}^2} \quad \checkmark\text{MA}$ $= 0,144... \text{ ℓ/m}^2$ Total amount of litres / <i>Totale aantal liters</i> $= 0,144 \times 3,96 \quad \checkmark\text{CA}$ $= 0,57 \text{ ℓ} \quad \checkmark\text{R}$ OR/OF Spread rate/ <i>Spreikoers</i> = $\frac{1 \text{ ℓ}}{6,9 \text{ m}^2} \quad \checkmark\text{MA}$ $= 0,144... \text{ ℓ/m}^2$ Paint needed for 1 coat/ <i>Verf nodig vir 1 laag</i> $= 0,144 \times 1,32 = 0,19... \text{ ℓ} \quad \checkmark\text{CA}$ Paint needed for 3 coats/ <i>Verf nodig vir 3 lae</i> $\checkmark\text{MA}$ $= 0,19... \times 3 \quad \checkmark\text{CA}$ $= 0,57 \text{ ℓ} \quad \checkmark\text{R}$	1MA multiplying with 3 1CA simplification 1MA dividing by 6,9 1CA simplification 1R rounding OR/OF 1MA dividing by 6,9 1MA adding the 2 coats and 1 1CA simplification 1CA simplification 1R rounding OR/OF 1MA multiplying with 3 1CA simplification 1MA dividing by 6,9 1CA simplification 1R rounding OR/OF 1MA dividing by 6,9 1CA simplification 1MA multiplying with 3 1CA simplification 1R rounding (5)	
3.3.4	$0,57 \text{ ℓ} \times 1\,000 \quad \checkmark\text{MCA}$ $= 570 \text{ ml} \quad \checkmark\text{CA}$ Not valid $\checkmark\text{O}$ <i>Nie geldig nie</i>	1MCA (from Q3.3.3 multiply by 1 000) 1CA simplification 1O verification	M L4

Q/V	Solution/Oplossing	Explanation/Verduideliking	T/L
	OR/OF 500 ml ÷ 1 000 ✓MCA = 0,5 l less than 0,57l ✓CA Tsidi's statement is invalid ✓O OR/OF 1 l covers/bedek 6,9 m² 500 ml covers/bedek $\frac{6,9}{2} = 3,45 \text{ m}^2$ ✓MCA Area to paint / Opp om te verf = 1,32 m² × 3 = 3,96 m² ✓CA The paint is not enough / invalid ✓O Die verf is nie genoeg / nie geldig OR/OF Coverage per coat/Dekking per laag = $\frac{500 \text{ ml}}{3} = \frac{0,5 \text{ l}}{3} = 0,166..$ ✓MCA Coverage /Dekking = 0,166 × 6,9 = 1,15 m² ✓CA 1,32 m² needs to be covered per coat/moet per laag gedek word. Not valid / Nie geldig nie ✓O	OR/OF 1MCA (from Q3.3.3 dividing by 1 000) 1CA simplification 1O verification OR/OF 1MCA area 1CA simplification 1O verification OR/OF 1MCA dividing 1CA simplification 1O verification (3)	
3.4.1*	Number of boxes/ Getal bokse = $\frac{162 \text{ cm}}{34,5 \text{ cm}}$ ✓MA ✓C = 4,695... ✓CA ∴ 4 boxes ✓R OR/OF Number of boxes/ Getal bokse = $\frac{1\,620 \text{ mm}}{345 \text{ mm}}$ ✓C ✓MA = 4,695... ✓CA ∴ 4 boxes ✓R	1MA dividing 1C conversion 1CA simplification 1R rounding down OR/OF 1C conversion 1MA dividing 1CA simplification 1R rounding down (4)	M L2 #

Q/V	Solution/Oplissing	Explanation/Verduideliking	T/L
3.4.2	Number of single files/ <i>Getal enkel lêers</i> $= \frac{162\text{cm}}{8,1\text{cm}} \quad \checkmark\text{MA}$ $= 20 \quad \checkmark\text{A}$ Number of files in boxes / <i>Getal lêers in 'n boks</i> $= 4 \times 4 \quad \checkmark\text{RT}$ $= 16 \quad \checkmark\text{CA}$ Difference in the number of files/ <i>Verskil in getal lêers</i> $= 20 - 16$ $= 4 \quad \checkmark\text{CA}$	CA number of boxes from 3.4.1 1MA dividing 1A simplification 1RT number of files in a box 1CA simplification 1CA difference in files (5)	M L3
3.4.3	Neater storage/ <i>Netjieser berging</i> $\checkmark\checkmark\text{O}$ OR/OF Files stand up straight/ <i>Die lêers staan regop</i> OR/OF Prevents dust on documents in the files/ <i>Verhoed dat stof op die dokumente in die lêers kom.</i> OR/OF It is easier to separate the files accordingly. <i>Dit is makliker om haar lêers te verdeel</i> OR/OF To categorise /organise her files/ <i>Dit is om haar lêers te katagoriseer /organiseer</i> OR/OF Prevent files from breaking/ damage/protect files <i>Verhoed dat lêers breek of beskadig/beskerm lêers</i>	2O reason (2)	M L4
3.4.4	$P = \frac{1}{16} \times 100\% \quad \checkmark\text{A}$ $= 6,25\% \quad \checkmark\text{MCA}$ $\quad \checkmark\text{CA}$	CA denominator from 3.4.2 1A numerator 1MCA denominator 1CA simplification (3)	P L2
		[35]	

QUESTION/VRAAG 4 [33 MARKS/PUNTE]			
Q/V	Solution/Oplossing	Explanation/Verduideliking	T&L
4.1.1	✓✓A Perennial garden bed./Meerjarige tuinbeddings OR/OF Compost / Kompos	2A correct feature (2)	MP L2
4.1.2	Water is scarce/Water is skaars OR/OF Rain water is free compared to tap water ✓✓O Reënwater is gratis in vergelyking met kraanwater OR/OF Pay less water bills/Betaal minder vir water OR/OF Water storage/ om water te stoor OR/OF To save water for future use Om water te spaar vir toekomstige gebruik OR/OF To harvest rain water Om reënwater op te gaar	2A Reason (2)	MP L4
4.1.3	Greenhouse roof/ gutters / Kweekhuis dak/geute ✓O OR/OF Livestock Barn roof/ gutters / Vee stoor dak/geute ✓O OR/OF Solar greenhouse roof / gutters/ Sonkrag kweekhuis	1A correct structure 1A 2nd correct structure Accept roof and gutter /pipe full marks (Any 2 structures) (2)	MP L4
4.1.4	Area/Oppervlakte = $\frac{1}{2} \times 17,024 \text{ m} \times 19,5 \text{ m}$ ✓RT ✓RT = 165,984 m² ✓CA	1RT correct height 1RT correct base 1CA area of a triangle NPR (3)	M L2
4.1.5	Option/Opsie A = R1 154 × 2 ✓MA = R2 308 ✓CA Option/Opsie B = R127,30 × 19 ✓MA = R2 418,70 ✓CA Option A. ✓O Opsie A.	1MA multiply by 2 1CA option A cost 1MA multiply by 19 1CA option B cost 1O best option (5)	MF L4

Q/V	Solution/Oplossing	Explanation/Verduideliking	T&L
4.2	Volume = $3,142 \times r^2 \times \text{height/hoogte}$ ✓SF $5000 \ell = 3,142 \times r^2 \times 220 \text{ cm}$ ✓C $5000\ 000 = 691,24 \times r^2$ $\frac{5\ 000\ 000}{691,24} = r^2$ ✓M $7233,377698 = r^2$ ✓S $\sqrt{7233,377698} = r$ ✓M $85,05 \text{ cm} = r$ ✓CA	1SF substituting 5000 1C converting ℓ to cm^3 1M dividing by 691,24 1S simplification 1M finding square root 1CA radius value NPR (6)	M L3
4.3.1*	$18 : 42$ ✓A $= 3 : 7$ ✓CA	1A correct order and values 1CA only if one value is correct or reversed order (2)	MP L1
4.3.2	Height/hoogte = $\frac{42''}{12''} = 3,5 \text{ feet/voet}$ ✓MA $3,28084 \text{ feet/voet} = 1\ 000 \text{ mm}$ $\therefore 3,5 \text{ feet/voet} = \frac{3,5}{3,28084} \times 1\ 000$ ✓C $= 1\ 066,799 \dots \text{mm}$ ✓CA OR/OF $3,28084 \text{ feet} = 1\ 000 \text{mm}$ 1 foot = n ✓MA $n = 304,79999 \text{mm}$ 1 foot = 12 inches Then 12 inches = $304,79999 \text{mm}$ 1 inch = $\frac{304,79999 \text{mm}}{12}$ ✓C $= 25,39999 \text{ mm}$ Therefore 42 inches = $42 \times 25,39999 \text{mm}$ $= 1066,7999 \text{mm}$ $= 1\ 066,8 \text{ mm}$ ✓CA	1MA converting to feet 1C converting to mm 1CA simplification OR/OF 1MA converting to feet 1C converting to mm 1CA simplification NPR (3)	M L2

Q/V	Solution/Oplossing	Explanation/Verduideliking	T&L
4.3.3	(a) iii ✓A (b) i ✓A (c) ii ✓A	3A correct Roman numeral (3)	MP L1
4.3.4	Q ✓✓A	2A correct letter (2)	MP L1
4.3.5*	✓✓A The notch labelled S is placed against B and the notch labelled R is placed against C ✓A <i>Die sitplek word bo-op die kantspanstukke geplaas</i> <i>Die uitkeping S word op B geplaas en die uitkeping R word teen C geplaas.</i>	2A mentioning the position of the 1st notch 1A second notch (3)	MP L4
		[33]	

Q/V	Solution/Oplossing	Explanation/Verduideliking	T&L
5.1.4	Number of litres in 23 gallons/<i>Getal liter in 23 gelling</i> $= 3,785 \times 23$ ✓ C $= 87,055$ litre ✓ S Cost of 87,055 litre/ <i>Prys vir 87,055 liter</i> $= 87,055 \times R15,97$ $= R1\ 390,27$ ✓ CA Valid/ <i>Geldig</i>. ✓ O OR/OF Number of litres / <i>Getal liter</i> $= \frac{R1400}{R\ 15,97}$ $= 87,664..$ litre ✓ S Number of gallons / <i>Getal gellings</i> $= \frac{87,664}{3,785}$ ✓ C $= 23,16$ gallons ✓ CA Can buy more with R1 400/<i>Kan meer koop met R 1400</i> Valid / <i>Geldig</i> ✓ O	1C gallons to litre 1S simplification 1CA cost of fuel 1O conclusion OR/OF 1S simplification 1C gallons to litre 1CA cost of fuel 1O conclusion NPR	MF L4 (4)

Q/V	Solution/Oplossing	Explanation/Verduidelik	T/L
5.1.5	1 full tank of fuel/ 1 vol tenk = 23 gallons /gelling ✓ A He can travel/ Hy kan reis = $23 \times 18 = 414$ miles Distance/afstand ✓ RT Greenfield - Fitchburg = 49 miles/myl Number of trips on 1 full tank /Getalritte met 1 voltenk $= \frac{414}{49} = 8,448..$ ✓ MA ✓ CA $\therefore$ 8 trips on 1 full tank / 8 ritte met 1 voltenk So, then he will fill tank back to 23 gallons Dan hervulhy die tenk tot 23 gelling Amount of fuel for 1 return trip/ brandstofvir 1 heen-en-weer reis $= \frac{98}{18} = 5,44$ gallon ✓ MA ✓ CA Left in a tank is $23 - 5,44 = 17,56$ gallons. ✓ MA ✓ CA Oor in die tenk is $23 - 5,44 = 17,56$ gelling OR/OF ✓ RT Distance/afstand_(Greenfield and Fitchburg) = 49 miles/myl Weekly must travel/ moet weekliks ry $= 5 \times 2 = 10$ trips ✓ MA He can travel = $23 \times 18 = 414$ miles with a full tank. ✓ MA Hy kan 414 myl ry met 'n vol tenk, 8 trips is $49 \times 8 = 392$ miles – now he needs to refill after Thursday's trips 8 ritte is 392 myl – dan hervul hy na Donderdag se terugkeer. With the full tank he only needs to travel Friday return trip / Hy ry danslegs Vrydag heen-en-weer ✓ A Friday trip: $49 \times 2 = 98$ miles / myl Used/Gebruik $= \frac{98}{18} = 5,44$ gallons/ gelling ✓ MA ✓ CA Left in a tank is $23 - 5,44 = 17,56$ gallons. ✓ MA ✓ CA Daar is $23 - 5,44 = 17,56$ gelling in die tenk oor	1A travel distance 1RT trip distance 1MA dividing 1CA number of trips 1MA dividing 1CA simplification 1MA subtracting 1CA simplification OR/OF 1RT trip distance 1MA weekly miles 1MA multiply 1A travel distance 1MA dividing 1CA usage on last day 1MA subtracting 1CA diff. between capacity and used gallons	M L3

Q/V	Solution/Oplossing	Explanation/Verduideliking	T/L
	18 miles on 1 gallon / 18 myl op 1 gelling ✓ RT 49 miles on x gallon / 49 myl op x gelling $x = \frac{48}{18} = 2,722\ldots$ gallon per trip / gelling per rit ✓ MA ✓ A Number of trips on 1st full tank / Getalritte met 1st voltenk $= \frac{23}{2,722\ldots} = 8,44\ldots$ ✓ CA ∴ 8 trips before he fills up again / 8 rittevoorhyweervolmaak ∴ 2 trips with second full tank/ 2 ritte met die 2de voltenk Fuel used / Brandstofverbruik ✓ MA ✓ CA $= 2,722\ldots \times 2 = 5,44\ldots$ gallon / gelling Left in the tank / Oor in die tenk ✓ MA ✓ CA $= 23 - 5,44\ldots = 17,56$ gallon / gelling. OR/OF Single Trip/Enkelrit = 49 miles /myl ✓ RT Number of gallons for 1 trip/ Getal gelling vir 1 rit ✓ MA ✓ A $= 49 \div 18 = 2,72$ Number of gallons for return trip/ virretoerrit $= 2,72 \times 2 = 5,44$ ✓ CA ✓ MA ✓ CA $23 \text{ gallons/gelling} \div 5,44 = 4,22 \text{ days/dae}$ $\approx 4 \text{ days/dae}$ No of gallons left / Hoeveelheid gelling oor ✓ MA ✓ CA $= 23 - 5,44 = 17,56$ gallons OR/OF	1RT trip distance 1MA dividing 1A travel distance 1CA number of trips 1MA multiplying 1CA simplification 1MA subtracting 1CA simplification OR/OF 1RT trip distance 1MA dividing 1A travel distance 1CA number of trips 1MA dividing 1CA simplification 1MA subtracting 1CA simplification OR/OF	

Q/V	Solution/Oplossing	Explanation/Verduideliking	T/L
	23 × 18 = 414 miles/myl ✓A Monday/Maandag : 49 × 2 = 98 miles/myl Tuesday/Dinsdag : 98 miles/myl Wednesday/Woensdag: 98 miles/myl Thursday/ Donderdag 98 miles/myl Totaal = 392 miles/myl ✓CA Fill up the tank on Thursday / <i>Maak die tenk vol petrol op Donderdag</i> Used per day /<i>Gebruik per dag</i> ✓ MA ✓ CA = 98 ÷ 18 = 5,44gallons Petrol left in tank/<i>Petrol oor in tenk</i> = 23 – 5,44 ✓ MA ✓ CA = 17,56 gallons Miles that can be travelled after Friday/<i>Myle wat nog gereis kan word na Vrydag</i> = 414 – 98 = 316 miles/myl Petrol left in tank/<i>Petrol oor in tenk</i> = 316 ÷ 18 = 17,56 gallons <td> 1A travel distance 1RT trip distance 1MA multiplying 1CA number of trips 1MA dividing 1CA simplification 1MA subtracting 1CA simplification (8) </td> <td></td>	1A travel distance 1RT trip distance 1MA multiplying 1CA number of trips 1MA dividing 1CA simplification 1MA subtracting 1CA simplification (8)	
5.2	$^{\circ}\text{C} = \frac{5}{9} (^{\circ}\text{F} - 32)$ $-7 = \frac{5}{9} (^{\circ}\text{F} - 32)$ $^{\circ}\text{F} = \frac{9}{5} \times -7 + 32$ = 19,4 ✓ CA ≈ 20⁰F ✓ R		